Software Manual (Generated from the source code with the help of ChatGPT)

Ratiometric Luminescence Thermometry Preprocessing and LIR Analysis Pipeline

Introduction

This software implements a complete batch-processing workflow for ratiometric luminescence thermometry. It imports Ocean Optics/Flame spectral files, removes cosmic ray artefacts, performs baseline correction, integrates two emission bands, computes the luminescence intensity ratio (LIR), and exports corrected spectra and summary statistics suitable for calibration and temperature analysis.

Software Architecture

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flame_data_loader.py
  ├─ load_folder()
  └─ load_single_file()

preprocess_spectra_compute_LIR.py
  ├─ remove_cosmic_rays()
  ├─ baseline_correct()
  ├─ integrate_region()
  ├─ export_corrected_spectrum()
  ├─ export_summary()
  └─ main()
```

Module: flame_data_loader.py

Purpose

Provides robust batch import of Ocean Optics/Flame text spectra.

Functions

load_folder()

Searches a directory for spectral text files, loads each spectrum using load_single_file(), and returns a dictionary containing wavelength and intensity arrays for every spectrum.

load_single_file()

Parses Ocean Optics text exports, locates the 'Begin Spectral Data' marker, extracts wavelength and intensity columns, ignores malformed rows, and converts the data into NumPy arrays.

Module: preprocess_spectra_compute_LIR.py

Purpose

Implements preprocessing, luminescence integration, LIR calculation, statistical summarisation and export.

Functions

remove_cosmic_rays()

Uses a local median and median absolute deviation (MAD) filter to detect and replace positive and negative cosmic-ray spikes while preserving genuine spectral features.

baseline_correct()

Computes the mean signal within a user-defined baseline region and subtracts it from the entire spectrum.

integrate_region()

Calculates the integrated emission intensity within user-defined wavelength regions using trapezoidal numerical integration.

export_corrected_spectrum()

Exports baseline-corrected spectra for archival, validation and downstream analysis.

export_summary()

Generates a summary table containing the mean and standard deviation of both emission-band integrals and the calculated LIR values across all processed spectra.

main()

Coordinates the complete workflow: imports spectra, removes cosmic rays, performs baseline correction, integrates the two luminescence bands, computes the luminescence intensity ratio, exports corrected spectra, and writes folder-level summary statistics.

Processing Workflow

- Batch import of Ocean Optics spectra.
- Cosmic-ray removal using local median/MAD filtering.
- Baseline estimation and subtraction.
- Integration of emission band L2.
- Integration of emission band L1.
- Calculation of the luminescence intensity ratio (LIR = L2/L1).
- Compilation of replicate statistics.
- Export of corrected spectra.
- Export of folder-level summary for thermometric calibration.

Input Parameters

The workflow exposes user-configurable parameters controlling the number of spectra processed, cosmic-ray filtering window and threshold, baseline region, integration limits for the two emission bands, and optional export of corrected spectra.

Outputs

- Corrected spectrum for each input file
- Folder-level summary containing mean and standard deviation of L1, L2 and LIR
- Console diagnostics reporting baseline values, cosmic-ray corrections and processed spectra

Quality Control

The pipeline validates input files, rejects malformed spectra, enforces odd filter-window sizes for MAD filtering, avoids division-by-zero during LIR calculation, and reports the number of cosmic-ray corrections applied to each spectrum.

Summary

This software provides a reproducible preprocessing workflow for ratiometric luminescence thermometry, combining robust spectral cleaning with quantitative emission-band integration and automated LIR computation suitable for temperature calibration experiments.